

If/Then Decisions

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Algorithms as Recipes

Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Categories and Labels

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Rules or Learned Patterns?

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Read four boundary phrases for Conditions and boundary cases. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Conditions and boundary cases. Proposed explanation: “The condition reads the starting value”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Transfer the method to a new display

A paper brightness display adds 3 to its starting number. If the current brightness is at most 7, subtract 2; otherwise add 1. Report and stop. Trace separate runs from starting values 3, 4 and 5. Identify the run that tests equality. Then show how its result changes if at most 7 is accidentally written as below 7.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A club admits a visitor if they have a ticket AND an accompanying adult. Test all four yes/no combinations. A student changes AND to OR. Which outcomes change?

6. Solve the given case

If a card has a star, send it to A; otherwise send it to B.

Sort a star, a circle, and a card with both shapes.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Use this exact rule: if a card has more than three dots, place it in tray A; otherwise use tray B. Trace cards with 2, 3 and 4 dots.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 10 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Add one transparent if/then branch and one else path to the Classroom Sorter.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A club admits a visitor if they have a ticket AND an accompanying adult. Test all four yes/no combinations. A student changes AND to OR. Which outcomes change?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Trace a decision from the current score

Use the paper-game procedure: add 2 to score; if score is at least 6 subtract 3, otherwise add 4; report and stop. Start at 2, then make a separate run starting at 5. For each run record the value at the condition, its true or false result, the branch and the final score.

My reasoning and evidence

13. W2 Read four boundary phrases

The known value at a condition is exactly 6. Decide whether each test is true or false: at least 6; above 6; at most 6; below 6. Explain which phrases include equality. Then name one value below and one value above 6 for neighbouring tests.

My reasoning and evidence

14. W3 Stop after the selected branch

In a separate decision the current score is 4. If score is at least 5 subtract 2; otherwise add 3; report and stop. A learner adds 3 to get 7, then subtracts 2 because 7 is now at least 5, and reports 5. Identify the extra action and give the correct trace and result.

My reasoning and evidence

15. W4 Transfer the method to a new display

A paper brightness display adds 3 to its starting number. If the current brightness is at most 7, subtract 2; otherwise add 1. Report and stop. Trace separate runs from starting values 3, 4 and 5. Identify the run that tests equality. Then show how its result changes if at most 7 is accidentally written as below 7.

My reasoning and evidence

16. W5 Handle an unknown input explicitly

The score is unreadable at the at-least-6 test. A helper says the test must be false and takes the otherwise branch. Explain what is unknown and why that choice is unjustified. State the next step required by this paper exercise and give two possible scores that would select different branches.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: If/Then Decisions

A fictional counter begins with a number, adds 2, then chooses one branch.

If the updated value is at least 6, subtract 1; otherwise double it.

Report once and stop. Use separate starting values 3,4,5.

Inclusive branching trace

	Start	After +2	At least 6?	Report
Run 1	3	5	false	10
Run 2	4	6	true	5
Run 3	5	7	true	6

Read a cell using both its row and column labels.

If/Then Decisions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Update first

Starts 3,4,5 become 5,6,7.



2. Test current value

$5 \geq 6$ is false; $6 \geq 6$ and $7 \geq 6$ are true.



3. Choose one branch

Double 5 to 10; subtract 1 from 6 to 5; subtract 1 from 7 to 6.



4. Report

The three independent runs report 10,5,6.

CHANGE ONE THING

Replace at least 6 with above 6.

Explain the changed result using the matching lesson.

Which input detects the difference between the two written conditions? Explain its entire run.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Include equality examples when reviewing sorter conditions, and trace from original input through output.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Include equality examples when reviewing sorter conditions, and trace from original input through output.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.
